

Section A — Concept Application

1.Question : Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

- CPU -> Pick based on workload & Check socket type.
- Motherboard -> Must match CPU socket + chipset. Ensure right form factor and features like (USB, networking).
- RAM -> Match motherboard type (DDR4/DDR5), speed, and slots. 16 GB is a good.
- Storage -> Confirm SATA or NVMe slots. Use SSD (fast boot/apps), add HDD if bulk storage needed.

➔ By Confirming compatibility we will see if they are working together or not :

- Matching CPU socket + chipset.
- Checking motherboard manual/QVL for RAM support.
- Verifying storage interface slots.
- Drivers are installed as per requirement.

2.Question : Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)	NVMe
Technology	Spinning magnetic platters + mechanical arm	NAND flash memory (no moving parts)	NAND flash memory via PCIe lanes
Speed	~80–160 MB/s	~500–600 MB/s	~2000–7000+ MB/s
Latency	High (mechanical delay)	Low	Ultra-low
Durability	Fragile (moving parts)	Durable	Durable
Cost per GB	Cheapest	Moderate	Highest
Best Use	Bulk storage, archives	Everyday OS/apps	High-performance workloads

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3.Question: Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.

Role of the GPU

- Graphics Processing Unit (GPU) handles renders images, videos, and visual effects.
- In sustained workloads (like office multitasking, video calls, light design work), it ensures smooth display output and offloads tasks from the CPU.
- A stable GPU prevents lag, screen tearing, and performance drops during prolonged use.

Role of the Cooling System

- Cooling system (fans, heatsinks, or liquid cooling) removes heat generated by the GPU and CPU.
- Maintains stable performance and extends hardware lifespan.
- Ensures the all work runs quietly and reliably under continuous load.

Why They Work Together

- GPU generates heat during sustained rendering.
- Cooling system dissipates that heat, keeping temperatures within safe limits.
- Together, they maintain performance + stability so the workstation can handle long office sessions without crashes or slowdowns.

4.Question: Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.

Hardware compatibility means ensuring that all computer components can work properly together. It is important because each component has specific requirements such as socket type, RAM type, connectors, power requirements, size, and supported interfaces.

Why It Matters

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- Ensures all components can be connected correctly.
- Prevents boot and compatibility errors.
- Provides stable and efficient performance.
- Prevents overheating and power-related problems.
- Helps avoid unnecessary hardware damage and expense.

Problems from Incompatible Components

- Computer may not boot.
- RAM, CPU, or storage may not be detected.
- System may crash or freeze.
- Components may overheat.
- PSU may cause power failure or instability.
- Components may not physically fit in the cabinet.
- BIOS/UEFI may not support the CPU.

5.Question: Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.

Choosing the PSU

- Calculate the total power consumption of CPU, GPU, motherboard, storage, fans, etc.
- Select a PSU with 20–30% extra wattage for future upgrades and power spikes.
- Check that it has the required 24-pin motherboard, CPU, GPU, and SATA power connectors..
- Ensure the PSU is compatible with the cabinet and power requirements.

Safety and Connection Checks

- Switch off and unplug the PSU before connecting components.
- Connect the 24-pin motherboard and 4/8-pin CPU power connectors properly.
- Connect the required PCIe power cables to the GPU.
- Connect SATA power to storage drives where required.
- Check that no cables are loose, damaged.
- Ensure RAM, GPU, and other components are fully intact
- Verify proper CPU cooling and fan connection.
- Perform a final inspection

6.Question: Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

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Checks for Successful POST

- Power on the computer and check that fans and LEDs work.
- Confirm that the display appears on the monitor.
- Check for POST beep codes or motherboard debug LEDs.
- Enter BIOS/UEFI and verify that CPU, RAM, storage, and other hardware are detected.
- Confirm that the system proceeds to the boot screen/operating system.

If POST Fails

- Turn off and unplug the computer.
- Check power cables and PSU connections.
- Reseat the RAM and graphics card.
- Check CPU and motherboard power connections.
- Disconnect unnecessary peripherals and try starting again.
- Check motherboard beep codes to identify the faulty component.
- Check BIOS/UEFI settings and update BIOS if CPU compatibility requires it.

Section B — Practical Coding Tasks

Task 1: Component Identification & Functions

Component	Function	Key Specification
Motherboard	Main circuit board that connects and allows communication between all hardware components.	Make/model visible on the motherboard
CPU (Processor)	Performs calculations and executes instructions; acts as the main processing unit.	Brand, model, cores/threads, clock speed
RAM	Temporary memory used to store data and programs currently being used by the CPU.	DDR4/DDR5, capacity (e.g., 8 GB/16 GB), speed
GPU	Processes graphics and sends visual output to the monitor.	Integrated or dedicated; model and VRAM

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Component	Function	Key Specification
Storage Device	Permanently stores the operating system, applications, and user data.	SSD/HDD, capacity, SATA/NVMe
Cooling System	Removes heat generated by the CPU and other components to maintain safe operating temperatures.	CPU fan, heatsink, liquid cooler, or case fans

- Motherboard – Connects CPU, RAM, storage, GPU, and other peripherals.
- CPU – Processes instructions and performs computational tasks.
- RAM – Provides fast temporary working memory.
- GPU – Handles graphics and video processing.
- Storage – Provides permanent data storage.
- Cooling System – Prevents overheating and maintains stable performance.

Task 2: Motherboard Ports & Socket Identification

Port / Header	Function
24-pin ATX Power	Supplies main power to the motherboard.
4/8-pin CPU Power	Provides power to the CPU.
SATA Ports	Connect SATA HDDs, SSDs, and optical drives.
USB Headers	Connect front-panel USB ports.
Front Panel Header	Connects power switch, reset switch, HDD LED, and power LED.
Fan Headers	Connect CPU and case fans.
Audio Header	Connects front-panel audio ports.
LAN/Ethernet Port	Provides wired network connectivity.
USB Ports	Connect external USB devices.
Audio Ports	Connect speakers, headphones, and microphones.
HDMI/DisplayPort	Provides display output when supported by the CPU/motherboard.

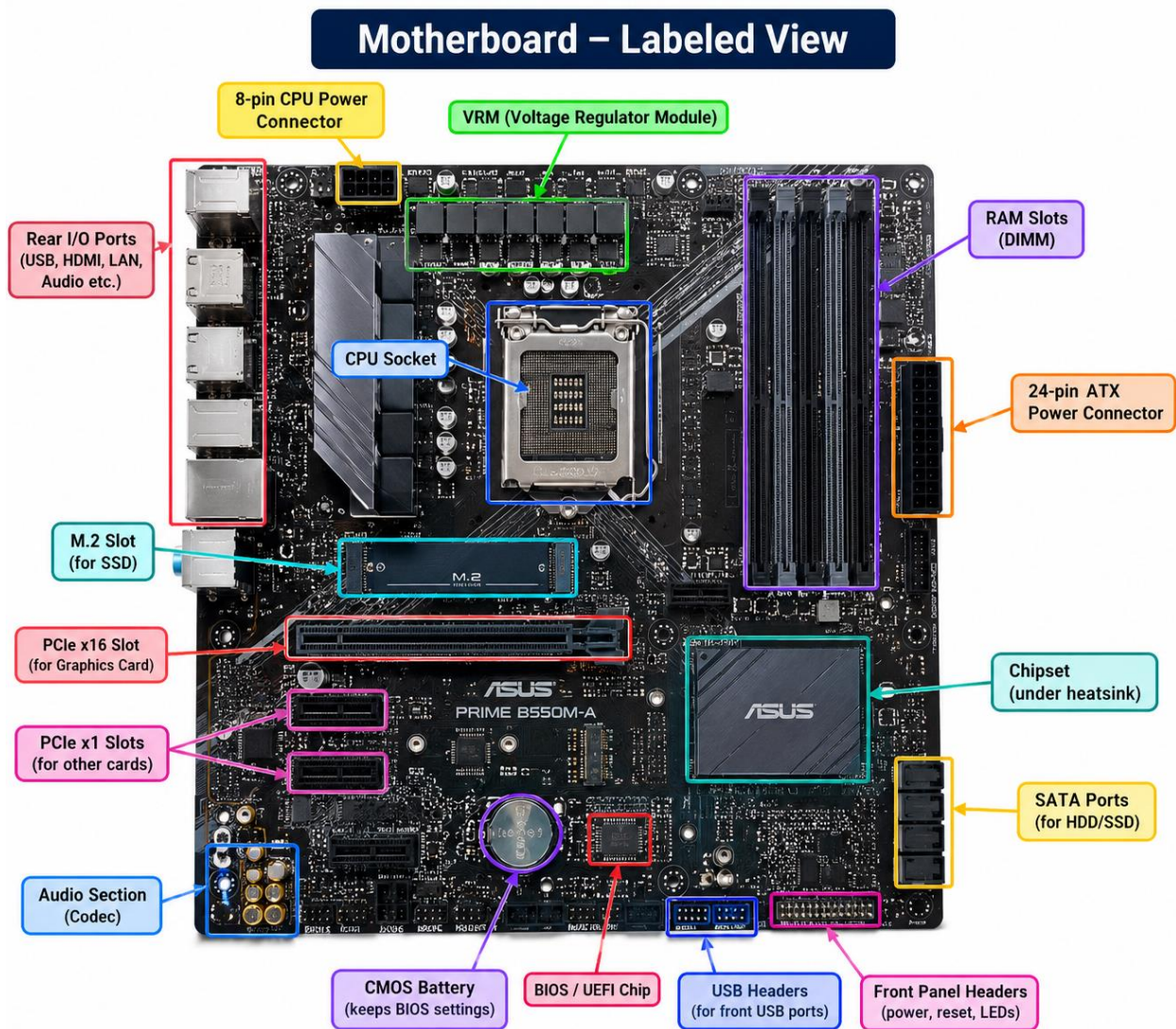
2. Expansion Slots

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Slot	Purpose
PCIe x16	Mainly used for dedicated graphics cards.
PCIe x1	Used for smaller expansion cards such as Wi-Fi, sound, or network cards.
M.2 Slot	Used for M.2 SSDs and, depending on the motherboard, Wi-Fi/Bluetooth modules.
DIMM Slots	Hold RAM modules.

3. CPU Socket

The CPU socket is the socket where the processor is installed. Its type must match the processor.



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Task 3: RAM & Storage Installation

. RAM Module Removal and Reinstallation

Procedure:

- Shut down the computer and disconnect the power cable.
- Open the cabinet and locate the RAM module.
- Push the locking clips on both sides of the RAM slot outward.
- Carefully remove the RAM module by holding its edges.
- Align the notch on the RAM with the slot.
- Press the RAM firmly into the slot until the locking clips click into place.
- Check that both ends are properly locked and the RAM is fully seated.

2. Storage Device Removal and Reinstallation

SATA Storage:

- Disconnect the SATA data and power cables.
- Remove the SSD/HDD from its mounting position.
- Reinstall the drive securely.
- Reconnect the SATA data and power cables.

NVMe Storage:

- Carefully remove the NVMe SSD.
- Insert it into the M.2 slot at an angle.
- Press it down and secure it with the screw.

3. ESD Precautions

- Turn off and unplug the system.
- Handle components by their edges.
- Avoid touching RAM contacts or SSD connectors.
- Use an anti-static wrist strap when available.
- Work on a clean, dry, non-static surface.

4. Verification

After installation:

1. Start the computer and enter BIOS/UEFI.
2. Confirm the RAM capacity is detected.
3. Confirm the SSD/HDD/NVMe drive is detected.
4. Boot into the operating system and verify the storage device appears correctly.

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Task 4: Full Desktop PC Assembly

Assembly Procedure

1. Install the CPU
 - Place the CPU into the correct motherboard socket.
 - Align the CPU marking with the socket marking.
 - Lock the CPU socket carefully.
2. Install CPU cooling
 - Apply thermal paste if required.
 - Mount the CPU cooler correctly.
 - Connect the CPU fan
3. Install RAM
 - Open the RAM slot clips.
 - Align the RAM notch with the slot.
 - Press the module until it clicks into place.
4. Install storage
 - Install the M.2/NVMe SSD into the M.2 slot and secure it, or
 - Install a SATA drive and connect SATA data and power cables.
5. Install the motherboard
 - Place the motherboard into the cabinet.
 - Secure it using the appropriate screws.
6. Install the GPU
 - Insert the graphics card into the appropriate PCIe x16 slot.
 - Secure it to the case.
 - Connect the required PCIe power cable.
7. Install the PSU
 - Mount the PSU securely in the cabinet.
 - Connect the 24-pin motherboard power cable.
 - Connect the 4/8-pin CPU power cable.
 - Connect GPU and storage power cables as required.
8. Final cable and safety check
 - Ensure all connectors are firmly seated.
 - Check that no cables are touching fans.
 - Check that there are no loose screws or metal objects inside the case.
 - Confirm the CPU cooler and fans are connected.

POST Verification

After completing the assembly:

- Connect the monitor, keyboard, and power cable.

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- Switch on the PSU and start the computer.
- Confirm that the monitor displays the startup screen.
- Check for beep codes or motherboard diagnostic LEDs.
- Enter BIOS/UEFI and verify that the CPU, RAM, storage, and GPU are detected.
- Confirm that the system proceeds to the operating-system boot screen.

SECTION C - Mini Capstone Project

- our project must: Identify the three hardware components you would check first when a newly assembled PC fails to boot, and explain why.
- Explain how motherboard–CPU compatibility affects system functionality, referencing the specific system you assembled or inspected.
- Describe how you would verify that RAM and storage devices are installed correctly, including what BIOS/POST should confirm.
- Produce a hardware-installation checklist of at least three steps to prevent similar failures in future deployments.
- Provide a technical justification for using SSDs instead of HDDs in modern office workstations, tying it back to the lab's use case.

1. First Hardware Components to Check

When a newly assembled PC fails to boot, the three most critical components to inspect are:

- Power Supply Unit (PSU)
 - Ensures stable power delivery. A loose 24-pin ATX or 8-pin CPU connector can prevent startup.
- CPU & Motherboard Connection
 - Incorrect seating or incompatible socket/chipset will stop POST.
- RAM Modules
 - Improperly seated or unsupported RAM prevents system initialization, often causing beep codes or no display.

These are foundational: without power, processing, and memory, the system cannot reach BIOS

2. Motherboard–CPU Compatibility

- Socket Type: Each CPU fits only specific sockets. Installing a mismatched CPU results in no boot.
- Chipset Support: Even with the correct socket, the motherboard chipset must support the CPU generation.
- BIOS Version: Some CPUs need a BIOS update before they can run.
- Impact: If compatibility fails, the system won't POST, and hardware detection errors appear.

3. Verifying RAM & Storage Installation

- RAM
 - Check modules are fully clicked into DIMM slots.
 - Use motherboard's recommended slot configuration
 - BIOS/POST should confirm correct capacity and frequency.
- Storage (SSD/HDD/NVMe)

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- Ensure SATA cables or M.2 drives are firmly connected.
- BIOS should list the drive under storage devices.

4. Hardware-Installation Checklist

To prevent failures in future deployments:

1. Match CPU socket and chipset before installation.
2. Reseat RAM and storage firmly, following motherboard slot guidelines.
3. Verify PSU connections (24-pin ATX, 8-pin CPU, GPU power if present).

5. SSD vs HDD Justification (Lab Use Case)

- Speed: SSDs (especially NVMe) deliver far faster boot and application load times than HDDs.
- Durability: No moving parts → less failure risk in a busy lab environment.
- Energy Efficiency: Lower power draw, reducing heat and noise.
- Office Lab Context: Students and staff benefit from quick logins, fast software launches, and reliable performance during sustained workloads. HDDs are better for bulk storage, but SSDs are optimal for everyday workstation use.

Final Note: By anchoring compatibility checks around CPU–motherboard pairing, verifying RAM/storage installation, and adopting SSDs for lab workstations, ABC Solutions can ensure smooth deployment and minimize boot-failure incidents.

Section D — AI-Augmented Learning

STEP 1 · BUILD WITH AI Use an AI tool of your choice (ChatGPT, Claude, Copilot, etc.) to help you produce a step-by-step troubleshooting checklist that:

- Guides a junior technician through diagnosing a desktop PC that fails to boot immediately after assembly, in the correct priority order.
- Covers at minimum the motherboard, RAM, storage, and PSU as possible failure points.
- Is presented as a clear checklist or simple decision flow a technician could follow on-site.
- Is applied to the ABC Solutions scenario from Section C to show which step would have caught the reported issue.

STEP 2 · TEST & DEBUG (WITHOUT AI)

- Then, working without AI, review the checklist and find at least one step that is missing, incorrectly ordered, or unsafe (for example, skipping ESD precautions or checking software before physical connections). Fix it yourself.

1. The exact prompt(s) you gave the AI tool

Mini Project: Workstation Deployment & Boot-Failure Diagnosis

Objective: ABC Solutions is setting up a new computer lab. After assembling several systems, some computers fail to start while others display hardware detection errors. Build a step-by-step troubleshooting checklist that:

- Guides a junior technician through diagnosing a desktop PC that fails to boot immediately after assembly, in the correct priority order.
- Covers at minimum the motherboard, RAM, storage, and PSU as possible failure points.
- Is presented as a clear checklist or simple decision flow a technician could follow on-site.
- Apply it to the ABC Solutions scenario to show which step would have caught the reported issue.

2. AI's Original Checklist:

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Check Power Supply Connections

- Ensure 24-pin ATX cable is fully seated
- Verify 8-pin CPU power connector is attached
- Check PSU switch is ON

Inspect Motherboard & CPU

CPU or motherboard incompatibility prevents POST.

- Confirm CPU is correctly seated in socket
- Verify thermal paste and cooler are installed
- Check motherboard supports CPU model

Reseat RAM Modules

Improperly seated RAM stops system initialization.

- Insert RAM firmly until clips lock
- Use recommended slots (A2/B2)
- Check BIOS/POST for correct capacity

Verify Storage Devices

Drives must be detected for boot sequence.

- Ensure SATA/NVMe drives are connected
- Confirm BIOS lists the drive
- Reseat cables or M.2 screws if missing

Run POST Diagnostics

Final Confirmation

POST confirms hardware detection before OS boot.

- Power on system
- Listen for beep codes
- Enter BIOS to verify CPU, RAM, and storage detection

3. Corrected Version (After Manual Review Without AI)

Observe ESD Precautions

Safety First

Prevent static discharge damage before touching components.

- Wear an anti-static wrist strap
- Work on a grounded surface
- Avoid carpeted areas

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Check Power Supply Connections

Stable power is required for startup.

- Ensure 24-pin ATX cable is fully seated
- Verify 8-pin CPU power connector is attached
- Check PSU switch is ON

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- Confirm CPU is correctly seated in socket
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Run POST Diagnostics

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4. Note Explaining the Change

- ✓ I added Step 1: Observe ESD Precautions at the beginning because the AI's original checklist skipped this critical safety measure.
- ✓ Handling components without grounding can damage hardware, making the troubleshooting unsafe.